

Semantic Shift Enhancement for Temporal Distribution Inference

Research Implementation

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1 Problem Statement

The base BPE/LP method achieves limited accuracy ($\log_{10}(\text{MSE})$ between -1.8 and -2.2) for temporal distribution inference. We propose semantic shift enhancement to capture word meaning evolution that BPE merge rules cannot detect.

Core Challenge: Words have different numbers of senses across time periods, requiring adaptive discovery rather than fixed assumptions.

2 Base Method Documentation: BPE/LP

2.1 BPE/LP (Byte Pair Encoding + Linear Programming) Methodology

The base BPE/LP method operates through the following pipeline:

Step 1: Merge Rule Extraction Extract ordered merge rules $m_1, m_2, \dots, m_T$ from tokenizer vocabulary construction.

Step 2: Token Pair Frequency Calculation Calculate token pair frequencies $c_{i,p}^{(t-1)}$ after applying merges m_1 through m_{t-1} .

Step 3: Linear Programming Dominance Constraints Build constraints ensuring chosen merges dominate alternatives:

$$\sum_i \alpha_i \cdot c_{i,m_t}^{(t-1)} \geq \sum_i \alpha_i \cdot c_{i,p}^{(t-1)} + \text{slack}_t, \quad \forall p \neq m_t \quad (1)$$

Step 4: Optimization Solve the linear programming problem:

$$\alpha_{BPLP} = \arg \min_{\alpha} \sum_{t=1}^T \text{slack}_t \quad (2)$$

subject to dominance constraints and $\sum_i \alpha_i = 1, \alpha_i \geq 0$.

Where $\alpha_{BPLP} = [\alpha_{1850s}, \dots, \alpha_{2020s}]$ represents the temporal distribution inferred by the base method.

3 Methodology: Clustering $\rightarrow$ Regression Pipeline

3.1 Step 1: Adaptive Sense Discovery with Hyperparameters

Problem: Cannot predefine number of word senses per decade.

Solution: Silhouette score optimization for K-means clustering with documented hyperparameters.

Algorithm:

1. Extract word contexts from decade-organized texts
2. Generate sentence embeddings using SentenceTransformer
3. Test K-means clustering for $K = 2, 3, 4, \dots, 8$
4. Calculate silhouette score with hyperparameters:

$$s(K; \theta) = \frac{1}{n} \sum_{i=1}^n \frac{b_i - a_i}{\max(a_i, b_i)} \quad (3)$$

where $\theta = \{\text{random_state} = 42, \text{n_init} = 10, \text{max_iter} = 300\}$

5. Select optimal K: $K^* = \arg \max_K s(K; \theta)$

3.2 Step 2: Feature Engineering - Fixed Vector Dimensions

Problem: Regression requires fixed-size numerical input features. Variable-length sense vectors break regression models.

Solution: Extract quantitative features with consistent dimensions using sense frequency averaging.

Sense Frequency Averaging (Critical Fix): Instead of concatenating individual sense frequencies (variable length), compute averaged sense frequencies:

$$\text{avg_sense_freq}_{\text{decade}} = \frac{1}{K^*} \sum_{k=1}^{K^*} \text{sense}_k \text{_freq}_{\text{decade}} \quad (4)$$

Fixed-Dimension Feature Vector:

$$\mathbf{x}_{\text{word}} = \begin{bmatrix} \text{silhouette_score} \\ \text{num_senses} \\ \text{cluster_separation} \\ \text{temporal_variance} \\ \text{normalized_inertia} \\ \text{word_freq}_{1850s}, \dots, \text{word_freq}_{2020s} \\ \text{avg_sense_freq}_{1850s}, \dots, \text{avg_sense_freq}_{2020s} \\ \text{entropy_change} \\ \text{context_diversity} \\ \text{syntactic_stability} \\ \text{frequency_acceleration} \\ \text{co_occurrence_shift} \\ \text{semantic_drift_velocity} \\ \text{cross_decade_overlap} \\ \text{cluster_stability} \end{bmatrix} \quad (5)$$

Feature Vector Dimensions: $\mathbf{x}_{\text{word}} \in \mathbb{R}^{49}$ (fixed for all words)

- Basic clustering features: 5 values (silhouette_score, num_senses, cluster_separation, temporal_variance, normalized_inertia)
- Word frequency per decade: 18 values (1850s-2020s)
- Average sense frequency per decade: 18 values (fixed length via averaging)
- Additional semantic features: 8 values (entropy_change, context_diversity, etc.)

Enhanced Feature Definitions:

- `silhouette_score`: Clustering quality measure (0-1)
- `num_senses`: Discovered sense count (K^*)
- `word_freq_{decade}`: Head word frequency per decade
- `avg_sense_freq_{decade}`: Average frequency across all discovered senses
- `cluster_separation`: Average inter-cluster distance
- `temporal_variance`: Variance in sense usage across decades
- `normalized_inertia`: Within-cluster sum of squares (normalized)
- `entropy_change`: Information entropy change over time
- `context_diversity`: Diversity of contexts per decade
- `syntactic_stability`: Consistency of syntactic roles

3.3 Step 3: Temporal Distribution Regression**Feature Aggregation Across Semantic Words**

To ensure statistical robustness while maintaining fixed dimensions, we aggregate features across all 50 semantic shift words:

$$\mathbf{x}_{avg} = \frac{1}{50} \sum_{w=1}^{50} \mathbf{x}_{word}^{(w)} \quad (6)$$

This produces a single averaged feature vector $\mathbf{x}_{avg} \in \mathbb{R}^{49}$ that represents the collective semantic behavior across all analyzed words.

Multi-Output Linear Regression Model

We employ standard multi-output linear regression to predict temporal distributions:

$$\boldsymbol{\alpha}_{predicted} = \mathbf{W} \cdot \mathbf{x}_{avg} + \mathbf{b} \quad (7)$$

Matrix Architecture:

- $\boldsymbol{\alpha}_{predicted} \in \mathbb{R}^{18}$ (temporal distribution across 18 decades)
- $\mathbf{W} \in \mathbb{R}^{18 \times 49}$ (weight matrix mapping features to decades)
- $\mathbf{x}_{avg} \in \mathbb{R}^{49}$ (averaged feature vector across 50 words)
- $\mathbf{b} \in \mathbb{R}^{18}$ (decade-specific bias terms)
- Computational verification: $(18 \times 49) \cdot (49 \times 1) + (18 \times 1) = (18 \times 1)$

Training Objective with Performance Monitoring:

$$\min_{\mathbf{W}, \mathbf{b}} \text{MSE} = \frac{1}{N} \sum_{i=1}^N \|\boldsymbol{\alpha}_{true}^{(i)} - (\mathbf{W} \cdot \mathbf{x}_{avg}^{(i)} + \mathbf{b})\|^2 \quad (8)$$

Where $\boldsymbol{\alpha}_{predicted} = [\alpha_{1850s}, \dots, \alpha_{2020s}]$ represents temporal distribution proportions.

Performance Evaluation Metrics: To ensure robust model assessment, we employ comprehensive error analysis:

$$\text{Training MSE} = \frac{1}{N} \sum_{i=1}^N \|\boldsymbol{\alpha}_{true}^{(i)} - \boldsymbol{\alpha}_{predicted}^{(i)}\|^2 \quad (9)$$

$$\text{Log}_{10}(\text{MSE}) = \text{log}_{10}(\text{Training MSE}) \quad (10)$$

Lower MSE values indicate superior model convergence and predictive accuracy.

3.4 Step 4: Feature Importance Analysis

Coefficient Analysis for Feature Selection: After regression training, analyze feature importance using learned coefficients:

$$\text{Importance}(f_j) = \frac{1}{18} \sum_{d=1}^{18} |\mathbf{W}_{j,d}|, \quad j = 1, \dots, 40 \quad (11)$$

Feature Selection Criterion:

$$\mathcal{F}_{\text{important}} = \{f_j : \text{Importance}(f_j) > \tau\} \quad (12)$$

where τ is a threshold (e.g., $\tau = 0.01$).

Features with coefficients close to zero indicate minimal contribution and can be removed.

3.5 Step 5: Integration with BPE/LP

Weighted Integration with Ablation Testing:

$$\boldsymbol{\alpha}_{\text{final}}(\lambda) = (1 - \lambda) \cdot \boldsymbol{\alpha}_{\text{BPLP}} + \lambda \cdot \boldsymbol{\alpha}_{\text{predicted}} \quad (13)$$

Lambda Ablation Study: Test integration weights: $\lambda \in \{0.0, 0.3, 0.7, 1.0\}$

- $\lambda = 0.0$: Pure BPE/LP method (no semantic enhancement)
- $\lambda = 0.3$: 30% semantic, 70% BPE/LP (proposed default)
- $\lambda = 0.7$: 70% semantic, 30% BPE/LP
- $\lambda = 1.0$: Pure semantic method (no BPE/LP)

Performance Evaluation:

$$\text{MSE}(\lambda) = \frac{1}{M} \sum_{i=1}^M \|\boldsymbol{\alpha}_{\text{ground_truth}}^{(i)} - \boldsymbol{\alpha}_{\text{final}}^{(i)}(\lambda)\|^2 \quad (14)$$

Select optimal $\lambda^* = \arg \min_{\lambda} \text{MSE}(\lambda)$.

With normalization constraint: $\sum_i \alpha_{\text{final},i} = 1$.

4 System-Level Validation Strategy

To evaluate the entire integrated pipeline rigorously, we employ a comprehensive train/test validation framework that addresses both individual component performance and overall system effectiveness.

4.1 Train/Test Data Split Protocol

The validation employs a standard 80/20 train/test split of the reference corpus, ensuring proper separation between training and evaluation data for both pipeline components.

Training Phase (80% Training Set):

BPE/LP Component Training:

- Calculate token pair frequencies using exclusively the 80% training corpus
- Generate merge rule dominance constraints from training data
- Solve linear programming optimization to obtain $\boldsymbol{\alpha}_{\text{BPLP}}$

Semantic Component Training:

1. Extract word contexts for 50 semantic shift words from training set only
2. Perform K-means clustering on training contexts to discover sense distributions
3. Generate feature vectors $\mathbf{x}_{word}^{(w)}$ for each word using training data
4. Construct proxy ground truth $\alpha_{true}^{(w)}$ from training set word frequencies
5. Average features and targets: $\mathbf{x}_{avg} = \frac{1}{50} \sum_{w=1}^{50} \mathbf{x}_{word}^{(w)}$, $\alpha_{avg} = \frac{1}{50} \sum_{w=1}^{50} \alpha_{true}^{(w)}$
6. Train regression model: $\alpha_{predicted} = \mathbf{W} \cdot \mathbf{x}_{avg} + \mathbf{b}$

Training Performance Metrics:

$$\text{Training MSE} = \|\alpha_{avg} - \alpha_{predicted}\|^2 \quad (15)$$

This metric assesses model fit quality on the training data.

4.2 Testing Phase (20% Hold-out Test Set)

Ground Truth Construction for Test Set: A comprehensive ground truth distribution $\alpha_{ground_truth_test}$ is constructed based on the known decade proportions of texts within the test set:

$$\alpha_{ground_truth_test,d} = \frac{\text{word count from decade } d \text{ in test set}}{\text{total word count in test set}} \quad (16)$$

BPE/LP Test Prediction: Using frequencies and constraints learned during training, generate α_{BPLP} prediction for the test distribution.

Semantic Model Test Prediction:

1. Extract contexts for 50 semantic shift words from test set only
2. Apply trained K-means models to assign sense labels to test contexts
3. Generate test feature vectors $\mathbf{x}_{word_test}^{(w)}$ using test data
4. Average test features: $\mathbf{x}_{avg_test} = \frac{1}{50} \sum_{w=1}^{50} \mathbf{x}_{word_test}^{(w)}$
5. Apply trained regression model: $\alpha_{predicted_test} = \mathbf{W} \cdot \mathbf{x}_{avg_test} + \mathbf{b}$

4.3 Integrated Model Evaluation

Lambda Ablation Study: The integrated prediction combines both components with varying weights:

$$\alpha_{final}(\lambda) = (1 - \lambda) \cdot \alpha_{BPLP} + \lambda \cdot \alpha_{predicted_test} \quad (17)$$

Comprehensive Performance Assessment: For each integration weight $\lambda \in \{0.0, 0.3, 0.7, 1.0\}$, calculate:

$$\text{Final Test MSE}(\lambda) = \|\alpha_{ground_truth_test} - \alpha_{final}(\lambda)\|^2 \quad (18)$$

Evaluation Interpretation:

- $\text{MSE}(\lambda = 0.0)$: Performance of BPE/LP method alone

- $\text{MSE}(\lambda = 1.0)$: Performance of semantic method alone
- $\text{MSE}(\lambda = 0.3)$ vs $\text{MSE}(\lambda = 0.0)$: Integration benefit quantification
- Optimal $\lambda^* = \arg \min_{\lambda} \text{MSE}(\lambda)$: Best integration weight

Success Criteria:

- Primary: $\min_{\lambda > 0} \text{MSE}(\lambda) < \text{MSE}(\lambda = 0.0)$ (integration improves base method)
- Secondary: $\text{MSE}(\lambda = 1.0) < \text{MSE}(\lambda = 0.0)$ (semantic method outperforms base method)
- Significance: Improvement margin $\geq 10\%$ to demonstrate practical benefit

This validation framework ensures rigorous evaluation of both individual components and their integration, providing clear metrics for assessing the semantic enhancement contribution to temporal distribution inference.

5 Implementation Requirements

5.1 Target Word Set (50 Semantic Shift Words)

Words selected across domains with documented semantic shifts:

- Computing: computer, memory, virus, web, mouse, terminal, browser, desktop, folder, window, icon, tablet
- Communication: broadcast, channel, network, wireless, stream, podcast, blog, post, tweet, feed
- Transportation: driver, traffic, platform, engine, hybrid, cruise, autopilot, terminal
- Finance: bank, credit, card, wallet, transfer, account, investment, portfolio
- Social: friend, profile, status, share, like, follow, community, connect
- Energy: power, battery, solar, grid, fuel, charge

5.2 Matrix Dimension Implementation Checklist

Critical Implementation Verification:

1. Ensure $\mathbf{x}_{word} \in \mathbb{R}^{40}$ for all words (fixed dimensions)
2. Implement averaging: $\mathbf{x}_{avg} = \frac{1}{50} \sum_{w=1}^{50} \mathbf{x}_{word}^{(w)}$
3. Verify $\mathbf{W} \in \mathbb{R}^{40 \times 18}$ (features $\times$ decades)
4. Confirm $\boldsymbol{\alpha}_{predicted} \in \mathbb{R}^{18}$ (decades)
5. Test matrix multiplication: $\mathbf{W}^T \cdot \mathbf{x}_{avg}$ produces valid result

5.3 Performance Reporting Requirements

Training Phase:

- Report MSE after each epoch
- Display feature coefficients and importance rankings
- Show convergence plots
- Calculate training time per iteration

Testing Phase:

- Compare MSE across different λ values
- Report improvement over base BPE/LP method
- Calculate confidence intervals for performance metrics
- Analyze feature stability across runs

6 Validation Framework

6.1 Hyperparameter Documentation

Clustering Hyperparameters:

- K range: $\{2, 3, 4, 5, 6, 7, 8\}$
- KMeans parameters: `random_state=42`, `n_init=10`, `max_iter=300`
- Silhouette threshold: $s_{min} = 0.3$ (minimum for reliable clusters)
- Embedding model: `SentenceTransformer('all-MiniLM-L6-v2')`

Regression Hyperparameters:

- Model: `LinearRegression()` with L2 regularization
- Feature scaling: `StandardScaler()`
- Train/test split: 80%/20%
- Cross-validation: 5-fold
- Regularization parameter: $\alpha_{reg} = 0.01$

6.2 Mathematical Validation

Computational Consistency Validation:

$$\text{SystemCheck}(\mathbf{W}, \mathbf{x}_{avg}) = \begin{cases} \text{VALID} & \text{if } \mathbf{W} \cdot \mathbf{x}_{avg} \in \mathbb{R}^{18} \text{ with } \mathbf{W} \in \mathbb{R}^{18 \times 49}, \mathbf{x}_{avg} \in \mathbb{R}^{49} \\ \text{INVALID} & \text{otherwise} \end{cases} \quad (19)$$

Feature Vector Validation:

$$\text{Valid}(\mathbf{x}_{word}) = \begin{cases} \text{TRUE} & \text{if } \dim(\mathbf{x}_{word}) = 49 \text{ for all words} \\ \text{FALSE} & \text{if } \dim(\mathbf{x}_{word}) \text{ varies} \end{cases} \quad (20)$$

6.3 Success Criteria

Mathematical Correctness:

- All matrix dimensions verify correctly
- Fixed-length feature vectors (no variable dimensions)
- Proper averaging implementation across 50 semantic words
- Hyperparameters documented in all equations

Performance Targets:

- Training MSE < base BPE/LP method MSE
- Feature importance analysis shows meaningful coefficients
- Integration improvement: 10-15% over base method via optimal λ^*
- Lambda testing demonstrates clear performance differences

Implementation Standards:

- MSE reported for all experiments
- Coefficient analysis included in all results
- BPLP method fully documented and integrated
- Ablation study completed across all λ values

7 Discussion and Next Steps

7.1 Immediate Implementation Priorities

1. Architectural Optimization

- Optimize ensemble feature aggregation methodology (Equation 6)
- Validate computational efficiency of matrix operations
- Establish robustness testing across multiple experimental runs

2. Feature Engineering Enhancements

- Replace variable-length sense vectors with averaged frequencies
- Add the 10 additional features specified in Equation (4)
- Implement feature importance analysis using Equation (9)

3. Performance Monitoring

- Add MSE reporting throughout training process
- Implement lambda ablation testing as in Equation (11)
- Create comprehensive performance comparison framework

7.2 Novel Contributions

- Presents mathematically rigorous integration framework for semantic enhancement with BPE/LP methodologies
- Introduces ensemble-based solution to variable-dimension feature challenges in temporal linguistics
- Establishes systematic feature importance methodology for temporal distribution inference
- Develops comprehensive ablation testing protocol for multi-method integration strategies